Realistic Mathematical Analysis of Modern Algorithms which Use SIMD Instructions for Better Performance

Supervisors: Prof. Cyril NICAUD
Dr. Pablo ROTONDO

Applicant: CHAU Dang Minh

Gustave Eiffel University, LIGM

Education and Research Experience

BEng in Computer Science *Univ. of Technology, Vietnam, 2024*

Relevant coursework: data structures and algorithms, and compilers.

MSc in Applied Mathematics *Univ. of Limoges, 2025*

Relevant coursework: stochastic optimization.

Internship: Compatibility between Open Automata (OA) and Session Types

Proved equivalence, mapped safe composition of OA to subtyping.

MSc in Mathematics & Computer Science *Gustave Eiffel Univ., in progress*

Relevant coursework: combinatorics, discrete optimization and complexity theory.

Internship: Complexity Analysis of the Regular Expression Matcher Hyperscan (by Intel)

Analyze FDR string matcher (SIMD extension of Shift-Or), correctness of decomposition-based matching, and longest literal in average.

Research Experience: Analysis of Algorithms

Example of a regular expression (regex): $(a + b \cdot c) \cdot d \cdot e^*$, equivalent to $(a \cdot d + b \cdot c \cdot d)e^*$.

Hyperscan executes as follows

1. extracts literals ($a \cdot d$ and $b \cdot c \cdot d$ in the example) from a regex,
2. matches all literals using FDR or another string matcher [1] *at once*,
3. verifies subregex matches using the previous results.

We assume a distribution over the regex trees, currently the BST-like family [2].

A useful statistics is the length of longest literals in average, which leads to analyzing the length Y_n of longest literals that an n-node subtree *yields* to its parent.

$$Y_{n+2} = \underbrace{I^{(n)}}_{\substack{\text{indicator of root} \\ \text{being concat.}}} \underbrace{(Y_{U_n} + Y_{n+1-U_n} + 1)}_{\substack{\text{take all of the subtrees,} \\ \text{plus the current}}} + \underbrace{J^{(n)}}_{\substack{\text{indicator of root} \\ \text{being alternation}}} \underbrace{\max(Y_{U_n}, Y_{n+1-U_n})}_{\substack{\text{take the longer} \\ \text{of the subtrees}}}, \quad Y_1 = Y_2 = 0.$$

[1] Qiu et al. (2021). Teddy: An efficient SIMD-based literal matching engine for scalable deep packet inspection.
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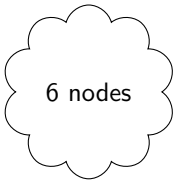

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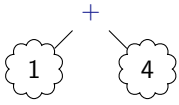

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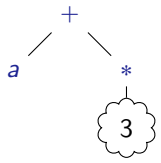

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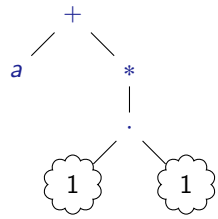

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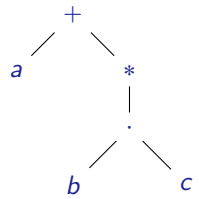

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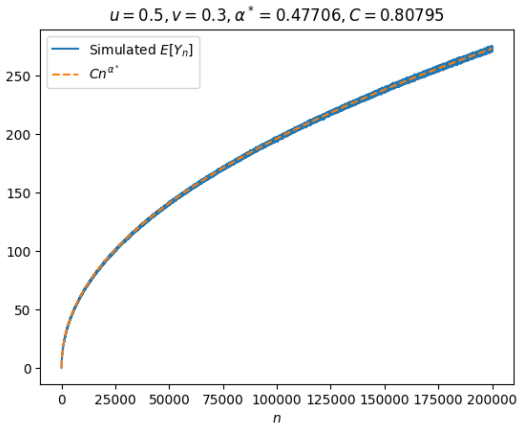
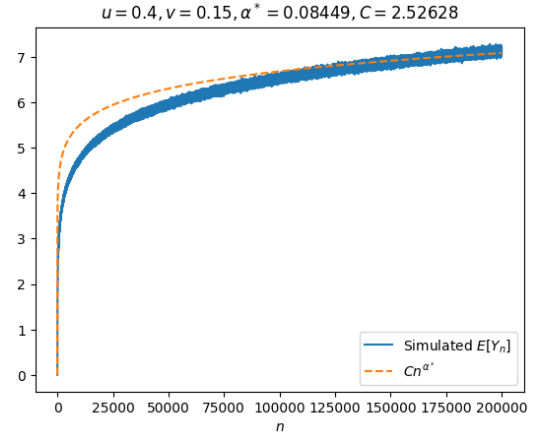
For $\mathbb{E}[Y_n]$ when $2\mathbb{E}[I^{(n)}] + \mathbb{E}[J^{(n)}] := 2u + v > 1$ (agreeing with real data),

Lower bound $\Omega(n^{\alpha})$ where α solves $\alpha + v2^{-\alpha} = 2(u+v) - 1$ | **Upper bound** $O(n^{2(u+v)-1})$

Conjecture: in some domain of (u, v), $\mathbb{E}[Y_n] \sim Cn^{\alpha^\star}$ for some $\alpha^\star \in (\alpha, 2(u+v)-1)$.





PhD Topic

Context and Motivation:

- Developers optimize their library algorithms in a fine-grained way e.g., they make assumptions on the input or leverage hardware features such as SIMD instructions.
- Researchers are interested in explaining if such optimizations are indeed efficient. Examples include both confirmation (TimSort [3]) and rejection (Lua's hybrid table [4])

Goal: Develop a framework for analyzing algorithms that use SIMD.

Fruitfulness:

- Direct applications to algorithm design.
- Requirements naturally lead to deep mathematical questions, e.g., $\mathbb{E}[Y_n]$ may not be analyzable by classical techniques.

[3] Auger, N., Jugé, V., Nicaud, C., & Pivoteau, C. (2018). On the worst-case complexity of TimSort.
[4] Martínez, C., Nicaud, C., & Rotondo, P. (2022). A probabilistic model revealing shortcomings in Lua's hybrid tables.

Professional Project

Long-term Goal: Pursue a career as a researcher/faculty member in theoretical CS.

Short-term Goal: Secure a postdoc fellowship.

Competencies to Build During the PhD:

- Acquire technical skills in analysis of algorithms.
- Learn how to communicate research effectively, both orally and in writing.
- Write and publish findings in respected conferences and journals.
- Learn French and gain university-level teaching experience.

Thank you for your attention !